INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

JUNEJA STREET

Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

Vihaan Juneja | Grade 9

Strategy development and research

Aarav Juneja | Grade 11

Risk analysis and execution

github.com/V-Juneja/ml-supertrend

Investment hypothesis and complete strategy

The strategy seeks short-term rebounds after unusually strong selling in volatile stocks. A fresh low alone is not a BUY: price must confirm an upward Supertrend reversal, recent oversold RSI and signal spacing. The machine-learning layer learns online from completed hypothetical long and short probes. Three model components use rolling outcomes, adverse-movement-weighted batches and an 8 by 8 market-context memory to adjust the Supertrend multiplier and ATR smoothing within bounds. The learning changes signal sensitivity and timing; explicit portfolio rules still set entry, position size and exit. The hypothesis fails when selling reflects lasting information, or costs erase any rebound.

Each probe opens at a candle open and matures at the close of its fifth candle. A move of one entry ATR gives the long probe a score of +1 and the matching short probe -1. Only matured outcomes can update later settings. Because probes overlap, their labels are dependent and are not account returns.

At equal confidence, one ATR of adverse movement halves a probe's batch weight. Context-memory influence rises with its supporting observations but is capped at 65%. A bounded update layer smooths parameter proposals; portfolio rules separately set entries, size and exits.

Why these stocks

DELL is the paper-trading example. The strategy scans liquid S&P 500 stocks; each BUY must pass liquidity and portfolio risk checks before an order is sized.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for liquid S&P 500 stocks. BUY opens one long; repeated BUYs do not add size. A confirmed SELL or protective stop closes it. Historical replay enters at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$
$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the multiplier. The boundaries sit mA above and below x; the baseline uses l = 24 and m = 1.4. They trail or reset around price. A close across the previous opposing band flips trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

The rolling learner uses up to 30 matured probes per direction: below 38% wins with mean score < 0 it proposes wider bands and longer ATR; above 62% with mean score > 0 it proposes narrower, faster settings. The batch learner weights each probe by confidence / (1 + adverse excursion in ATR units), then summarizes normalized gain, profit factor and downside. A 1-ATR adverse move halves its weight.

Context memory conditions probe outcomes on market state. Its 0–1 trend score combines 35% inverse 64-candle entropy, 35% scaled 64-candle Hurst persistence and 30% normalized ADX(14); 0 is choppy, 1 trending. Relative volatility is current ATR / trailing 100-candle mean, scaled 0–1 at 2.5×. The two axes index 8×8 = 64 cells. Each cell exponentially smooths probe gain, downside and parameter proposals (50-update half-life). A Gaussian-weighted 3×3 neighborhood forms its local proposal; sample count sets the local/global blend, capped at 65%. This adapts indicator settings, not capital allocation.

The combiner blends rolling and context proposals with available batch feedback. Larger multiplier requires a larger reversal; longer ATR responds more slowly. Deadbands, update caps, parameter bounds and baseline reversion limit drift. Separate portfolio rules control capital.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current multiplier or ATR length; theta tilde is the rounded, bounded target; alpha is the update fraction. For multiplier 1.40, target 1.50 and alpha 0.20, the update is 1.42. Deadbands ignore tiny changes; caps constrain each update and periodic reversion pulls settings toward baseline. The multiplier stays 0.5-5.0 and ATR length 5-100.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within the prior 50 candles and at least 10 candles since the last signal. RSI compares Wilder-smoothed gains and losses over 14 candles on a 0–100 scale. SELL requires a fresh-high reversal, downward flip and recent RSI above 70. Signals use completed candles; volume and tick-pressure filters are off.

Implementation check. Yahoo data: 780 DELL bars, 776 matured probes per side; multiplier 1.40 to 2.145, ATR stayed 24; 6 BUY / 8 SELL. Prefix outputs matched.

2 ML Supertrend replay and backtest charts

We replayed ML Supertrend BUY emails from 8 September 2025 to 14 October 2026, carrying the last list through 30 October (290 sessions). Each list was applied at the next session open, starting with $100,000 and charging 10 basis points per side. Historical sizing allowed up to ten holdings; the current declaration caps the portfolio at five.

ML Supertrend replay result	Value
Ending marked equity	$151,542.68
Total return / annualized growth	+51.54% / +43.92%
Maximum drawdown	-21.10%
QQQ benchmark return	+19.39%

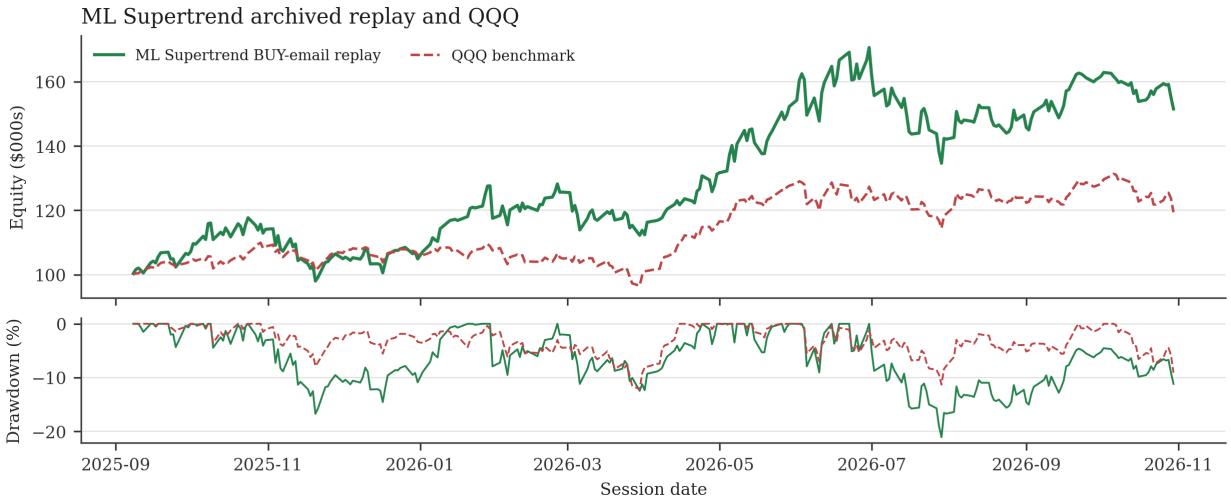


Figure 1. Ending equity was $151.5k versus QQQ at $119.4k; the 21.10% maximum drawdown shows returns came with sharp setbacks. Data: Alpaca SIP split-adjusted bars; our replay.

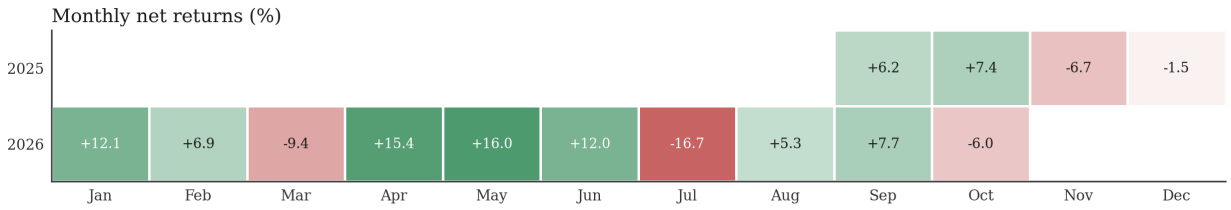


Figure 2. Nine of 14 months were positive; the largest monthly loss was 16.7% in July 2026. Monthly results inform position sizing and drawdown limits. Data: same replay.



Figure 3. The replay returned +51.54% under the base cost assumption. Return fell to +39.11% at 20 bps per side and +43.49% with a one-session delay, showing sensitivity to execution assumptions. Green bars meet or exceed the base result; red bars fall below it.

All 1,254 fills and 290 equity marks reproduce against the saved replay. Replay method · GitHub repository.

3 Trade expectancy and risk tests

Net Trade Expectancy Equation

$$e = pW - (1-p)D, \qquad p_{\text{BE}} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

Trade statistic	Historical result
Completed trades	622
Win rate	46.62%
Mean net profit / return per trade	+$81.73 / +0.8173%
Median trade return	-0.6268%
Average net winning / losing return	+8.05% / -5.50%
Profit factor	1.28
Break-even win rate at observed payoffs	40.59%
Best / worst net trade return	+62.80% / -40.03%
Longest losing streak	15 trades

These 622 closed trades come from the ML Supertrend email replay. The 46.62% win rate exceeded its 40.59% break-even level; profit factor was 1.28. The median trade lost 0.6268%, while larger gains lifted the mean. Open positions are excluded. Trade calculations and fills: GitHub trade evidence.

ML Supertrend one-year Monte Carlo

We resampled sequential equity changes from the archived ML Supertrend replay (290 sessions, 8 September 2025–30 October 2026) in circular five-session blocks. Each of 50,000 paths spans 252 sessions. Source, calculation and chart: Monte Carlo files.

One-year path statistic	Result
Median terminal return	+43.98%
5th / 95th percentile terminal return	-11.85% / +131.43%
Paths ending below start	10.87%
Mean return in worst 5% of paths	-21.93%
95th percentile maximum drawdown	35.46%

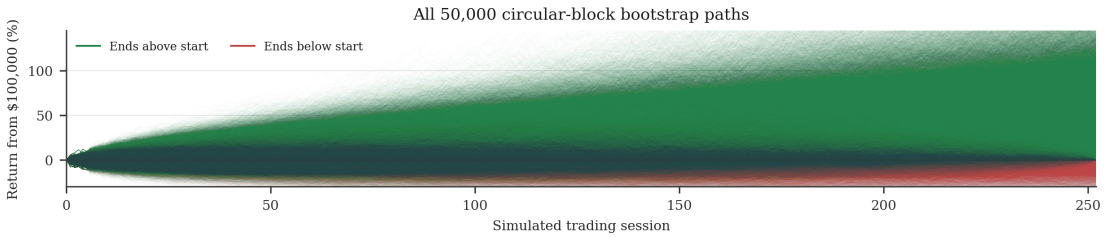


Figure 4. All 50,000 resampled paths are shown: green ended above the start; red below. Although the median one-year return was +43.98%, 10.87% lost money; the 5th-percentile return was -11.85%. These scenarios show a wide range of outcomes, not a single expected path.

4 Portfolio construction and risk

False reversals can cluster, overnight gaps can cross stops and technology names can fall together. Entry spacing, stop-risk budgets, concentration caps and account-loss rules address these strategy-specific risks. Limits are declared choices, not optimized. Risk-test assumptions and code.

Control	Execution rule and purpose
Exposure	Long-only, no borrowing; at most five names, 15% of equity per name and 75% gross stocks. Five names at the individual cap use the full portfolio allowance.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Execution integrity	Use one account; reject duplicate BUY entries. Log accepted/rejected signals, settings, quotes, order and fill times, fees, stop and exit reason. Freeze the declared rules; close by the competition cutoff.

Position Sizing Equation

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.15E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At the recorded DELL close of $563.17, a $15,000 cap permits 26 shares ($14,642.42). The $500 loss budget in a $100,000 account allows at most $19.23 stop distance at that size. A wider two-ATR stop reduces shares. The 15% name limit and five-position limit allow at most 75% gross exposure; remaining equity stays in cash. Position size can be lower when the stop-risk budget binds.

Gap and execution stress

Stop execution stress	Loss / planned stop loss	Account loss at 2% risk budget
Fills at intended stop	1.0 x	2%
Loss exceeds stop distance by 50%	1.5 x	3%
Loss doubles the intended distance	2.0 x	4%
Gap produces three times planned loss	3.0 x	6%

The table multiplies planned account risk by the realized stop-distance overrun. A gap can exceed every scenario shown, so the 6% halt trigger cannot cap losses.

Stress grid: 96 combinations of horizon, exposure, volatility and correlation (100,000 paths each); daily Gaussian, t(5), gap and liquidation tests use 40,000 paths per model. At the 75% ceiling, common 10%/20% price declines produce 7.5%/15% account losses before costs. Opening gaps can exceed the 6% halt threshold.

Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold at most five names, 15% per name and 75% total stocks. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.